



SUMMARY

4+ years building scalable AI-based systems, architecting data science models, and wrangling Big Data with Python, SQL, and cloud platforms GCP. Passionate about AI research and empowering businesses with actionable insights and fostering a data-driven culture.

SKILLS

Natural Languages: Arabic – Native / English – Fluent / French – Fluent / Amazigh - Fluent
Programming Languages: Python, Java, SQL, R, C#, BASH shell scripting.
Big Data Technologies: Hadoop, Spark, Scala, Databricks, Apache Kafka, Apache Airflow.
Data Storage and Querying: MySQL, Teradata, MS SQL Server, MongoDB, Firebase, Cloud Storage, S3.
Cloud Platforms: GCP (BigQuery, Cloud Storage, Dataflow, Pub/Sub, Looker, Artifact Registry, Cloud Build, Cloud Run, Compute Engine), AWS (S3, EC2, Lambda, RDS, Glue, API Gateway, SageMaker), Virtual Private Cloud, Dedicated Server.
Containerization & Orchestration: Docker, Kubernetes
Data Science and Miscellaneous Technologies: ETL, Data Science Pipeline, Data Modeling (entity-relationship diagrams (ERDs), Data modeling languages(UML), Data modeling software (Lucidchart, draw.io)), Statistics, Time Series Analysis, A/B Testing, OOP, APIs, Excel, Git, Power BI.

EXPERIENCE

 Cloud Data Engineer
July 2024 - Present

- **Data Migration from Oracle DB to GCP:** Building reliable and efficient pipelines to perform bulk data migration from legacy Oracle DBs to GCP BigQuery, leveraging GCP Dataflow and Apache Beam SDK. Also architecting the general architecture of the cloud end-to-end project
- **Operations and APIs migration from op-premise systems to GCP:** Intensive Java code refactoring to migrate the APIs and all operations from legacy on-premise systems to GCP using SpringBoot and SpringBatch.

Skills: ETL and ELT operations, Apache Kafka, Java (SpringBoot and SpringBatch), Python, Oracle PL/SQL, On Premise and Cloud Data Warehousing, Encryption and Hashing, GCP(BigQuery, Dataflow, Cloud Monitor, Cloud Build, Pub/Sub, Compute Engine, GKE), CI/CD, Agile JIRA & ITIL Confluence, Terraform, Elastic-Search

 AI & Data Consultant
December 2022 - July 2024

- **E-MOULOUYA Environmental Wildlife Monitoring and MLOps :** Research and Development of the E-Moulouya AI-based mega project (First Prize Seoul Smart City, Second Prize IEEE Smart City 2023) ([paper](#)). using computer vision to achieve a high accuracy detection, tracking and classification. System design and deployment leveraging GCP Pub/Sub for stream processing of bird recognition data. Training the Deep Learning models in GCP (Google Colab and Vertex AI)
- **Smart Traffic Management:** Estimation of traffic line length using GMM and Object Detection to ease congestion. Classification of vehicles to gather statistical data for the smart city.
- **AI Chatbot:** Development of an Arabic AI chatbot using Jais and RAG to act as an administrative assistant for the people of the city.
- **CI/CD Operations and MLOps for R&D Team:** automate operations of AI research and development CI/CD by using auto trigger cloud function, Github, cloud Build and Vertex AI.
- **IoT Digital Twin:** Setup and consolidate IoT data, streaming it to GCP through Pub/Sub, Applying transformations with Dataflow, and saving it the data in BigQuery.
- **Enterprise Scale Attendance and Employees Management System:** Assisting in decision making for stakeholders by incorporating a dashboard with multiple KPI's for visualizing and tracking the employees of BeAt Berkane using PowerBI. Implementation of a novel and optimized Face recognition ([paper](#)) system as an API using Django and database design, while securing it against attacks by using different encryption/Hashing methods.

Skills: ETL and ELT operations, Apache Kafka, FastAPI, MySQL, Firebase, IoT, On Premise and Cloud Data Warehousing, Encryption and Hashing, Face Recognition, Apache Airflow, Apache Kafka, GCP(BigQuery, Dataflow, Cloud Monitor, Cloud Build, Pub/Sub, Compute Engine, GKE, Vertex AI), CI/CD, Computer Vision, XGBoost, Databricks, Agile JIRA & ITIL best practices environment, Scientific Paper Writing.

 Data Scientist
Jan 2022 - December 2022

- Building Data Exploration and cleaning pipelines by collaborating with data analysts and data modeling of sport events for Genius Sports.
- Automating reports generation of data visualization resulting into 50% faster insights reporting and decision making
- Employing Correlation, Clustering and Ensemble Trees to classify and group events together as an innovative approach to deal with clusters of data and suggest KPI of sports events related data.
- Migrating data to the cloud from Teradata to GCP BigQuery.

Skills: GCP BigQuery, GCP Dataflow, GCP IAM, Data Analysis, SQL Teradata, BI, Data visualization, Data Warehousing, Excel, Python, AWS Authentication, AWS Lambda, AWS API Gateway, Amazon S3.

 Data Scientist
April 2020 - Jan 2023

- Developed data models for a variety of clients using SQL and data modeling methods.

- Implemented machine learning models to solve a variety of problems, such as fraud detection, customer segmentation, and product recommendation.
- Developed E-Commerce businesses solutions in a data intensive reliable architecture and set of technologies using Java SpringBoot and Python FastAPI.
- Developed mobile apps for Gym membership management and sports activities using Java Android Studio.
- Designed and implemented data pipelines to collect, transform, and load data from various sources using on-premise technologies (Apache Kafka, Talend) & Cloud Technologies (GCP Pub/Sub, Dataflow).
- Worked with business analysts to understand requirements and translate them into data models.
- Collaborated with team members to complete projects using JIRA and TRELLO.

Skills: Data modeling, Data Warehousing, Data analysis, Data visualization, Business intelligence, Problem solving, Communication, Java SpringBoot, Android Studio, Python, FastAPI, MySQL database, Oracle database, Hadoop, Teradata, Google BigQuery, SQL, PowerBi, Looker, Project management tools(JIRA Scrum, Trello).

Big Data Developer

Jun 2021 - Aug 2021

- Developed a web application for dashboard visualisations of employees data for the council. Resulting into ease of access to information and gaining new insights into the work culture inside, using **Scala** to handle and query on the massive amounts of data, **R** initially for visualization, and later **Python** by PySpark.

EDUCATION

MSc in Advanced Information Systems, École Supérieure de Management, d’informatique et de Télécommunication

BSc in Business Intelligence, École Supérieure de Technologie, Oujda

PUBLICATIONS

28th IEEE ISCC 2023 Aggregating Multiple Embeddings: A Novel Approach to Enhance Reliability and Reduce Complexity in Facial Recognition *Houssam Benaboud, **Walid AMARA**, Amal Ezzouhri, Fatima Eljaïmi, Wiam Rabhi, and Zakaria Charouh*

•**28th IEEE ISCC 2023** An Integrated Framework for Bird Recognition Using Dynamic Machine Learning-Based Classification *Wiam Rabhi, Fatima Eljaïmi, **Walid AMARA**, Houssam Benaboud, and Zakaria Charouh*

•**IEEE DataPort** LaSBiRD: Large Scale Bird Recognition Dataset

•**IEEE DataPort** E-Moulouya BDD: Extended Moulouya Bird Detection Dataset

CERTIFICATIONS



Databricks Apache Spark SQL



Google BI Professional



ETL & Data Pipelines, Airflow & Kafka



IBM DevOps



Introduction to Agile Development & Scrum



Kaggle Data Courses



IBM Exploratory Data Analysis for ML



Deep Learning & ML Specialization

WRITTEN WORKS AND SPEAKING EXPERIENCE



Winning the Seoul Smart City Prize 2023

Thanks to our efforts and dedication, Majal Berkane was able to win the first prize at seoul smart city, as we presented our project of E-Moulouya that I collaborated on, this project holds these major components

- Cutting Edge AI
- Real-Time Computer Vision
- Edge Computing
- System Architectural Design
- Green & Environmental Friendly Solution.



Second Prize at IEEE Smart City Romania

As we presented our stack of cutting edge smart city solutions, we had the opportunity to engage in a challenge against giant cities around the world, in which we succeeded in proving that we can deliver cutting edge tech at any environment.

HOBBIES AND INTERESTS

[Youtube Channel](#) and [Instagram Page](#) for AI content and knowledge sharing. Keeping up with Scientific Research, and taking good care of my health by working out, and having quality time with friends and family.